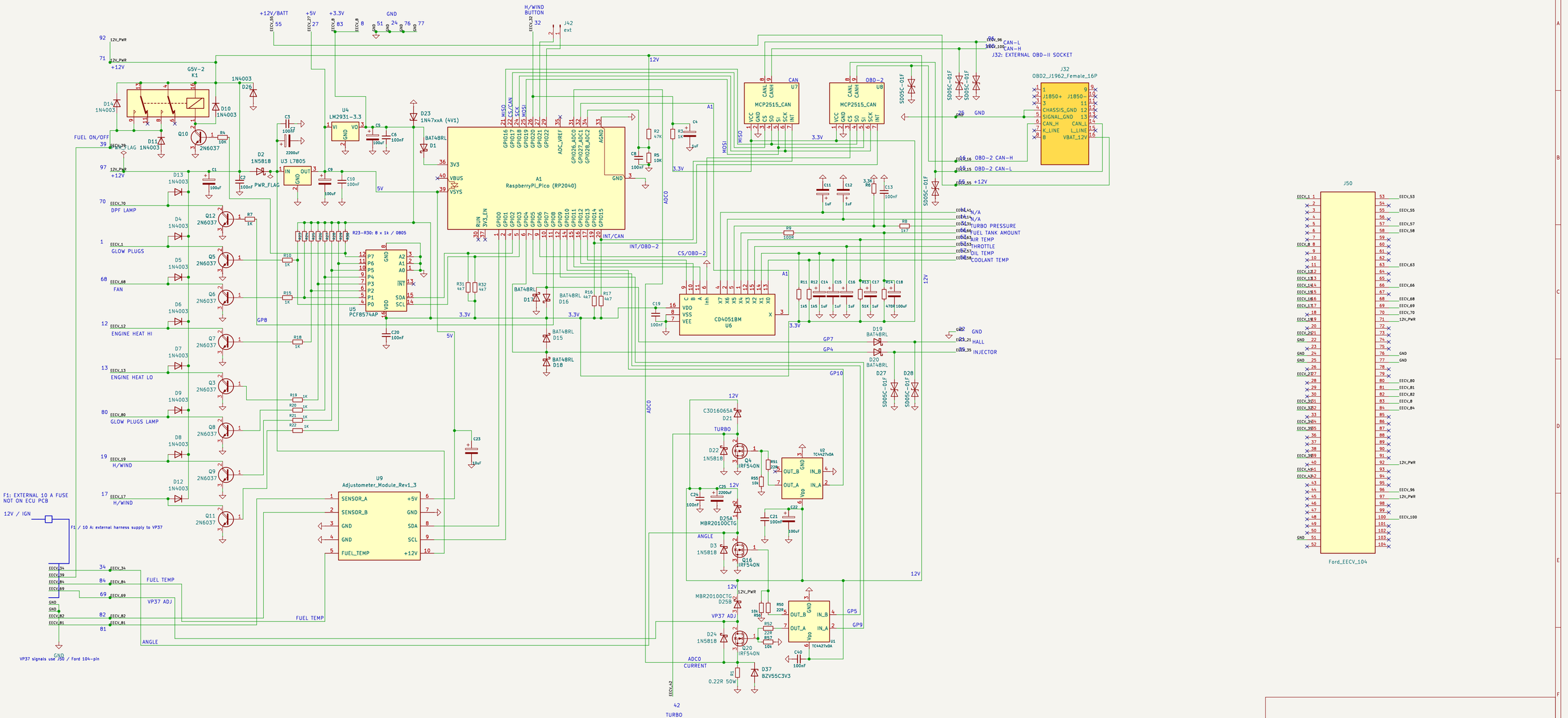

+12V/BATT
+5V
+3.3V
GND
H/WIND BUTTON
J42 ext
12V_PWR
G5V-2 K1
1N4003
FUEL ON/OFF
ECCV_39
VP37_FLAG
Q10 2N6037
R4 10K
D11 1N4003
D16 1N4003
D10 1N4003
D26
C3 100nF
C7 2200uF
U4 LM2931-3.3
VI
VO
GND
C5 100nF
C6 100uF
D23 1N47xxA (4V1)
BAT48RL D1
D2 1N5818
PWR_FLAG
U3 L7805
IN
OUT
C1 100uF
C2 100nF
C9 100uF
C10 100nF
5V
A1 RaspberryPi_Pico (RP2040)
3V3
VBUS
VSYS
RUN
3V3_EN
ADC_VREF
AGND
R2 47K
R5 10K
C8 100nF
ADCO
MISO
MOSI
SCK
CAN_CS
CAN_INT
INT/CAN
INT/OBD-2
CS/OBD-2
U7 MCP2515_CAN
U8 MCP2515_CAN
OBD-2
C4 1uF
R3 1K
SD05C-01F
J32 OBD2_J1962_Female_16P
J1850+
J1850-
CHASSIS_GND
SIGNAL_GND
CAN_H
CAN_L
K_LINE
L_LINE
VBAT_12V
CAN-L
CAN-H
J32: EXTERNAL OBD-II SOCKET
OBD-2 CAN-H
OBD-2 CAN-L
+12V
C11 1uF
C12 1uF
3.3K R6
C13 100nF
R8 1K7
R9 100R
N/A
N/A
TURBO PRESSURE
FUEL TANK AMOUNT
AIR TEMP
THROTTLE
OIL TEMP
COOLANT TEMP
U6 CD4051BM
C19 100nF
C11 1uF
C12 1uF
C15 1uF
C16 1uF
C17 1uF
C18 100uF
R11 1k5
R12 1k5
R13 51K
R14 470R
D19 BAT48RL
D20 BAT48RL
GP7
GP4
GP10
HALL
INJECTOR
D27 SD05C-01F
D28 SD05C-01F
J50
Ford_EECV_104
R23–R30: 8 x 1k / 0805
U5 PCF8574AP
SDA
SCL
INT
C20 100nF
R31 4k7
R32 4k7
BAT48RL D14
BAT48RL D15
BAT48RL D16
BAT48RL D18
R16 4k7
R17 4k7
C23 10uF
R10 1K
R15 1K
R18 1K
R19 1K
R20 1K
R21 1K
R22 1K
GP8
DPF LAMP
GLOW PLUGS
FAN
ENGINE HEAT HI
ENGINE HEAT LO
GLOW PLUGS LAMP
H/WIND
H/WIND
Q12 2N6037
Q5 2N6037
Q6 2N6037
Q7 2N6037
Q3 2N6037
Q8 2N6037
Q9 2N6037
Q11 2N6037
D4 1N4003
D5 1N4003
D6 1N4003
D7 1N4003
D9 1N4003
D8 1N4003
D12 1N4003
D13 1N4003
R7 1K
U9 Adjustometer_Module_Rev1_3
SENSOR_A
SENSOR_B
FUEL_TEMP
+5V
SDA
SCL
+12V
F1: EXTERNAL 10 A FUSE NOT ON ECU PCB
12V / IGN
F1 / 10 A: external harness supply to VP37
VP37 signals use J50 / Ford 104-pin
FUEL TEMP
VP37 ADJ
ANGLE
TURBO
ADCO CURRENT
0.22R 50W
R1
D37 BZV55C3V3
C3D16065A D21
D22 1N5818
Q4 IRF540N
D25A MBR20100CTG
D3 1N5818
Q16 IRF540N
D25B MBR20100CTG
D24 1N5818
Q20 IRF540N
U2 TC4427xOA
U1 TC4427xOA
R51 22R
R55 10k
R50 22R
R52 22R
R57 10k
C25 100nF
C26 2200uF
C21 100nF
C22 100uF
C40 100nF
12V_PWR
PCB1: R / ceramic C = 0805; electrolytics = radial THT / 25V
CAN and R1: measured footprints; R1 mounting follows accepted mechanical baseline
TC4427: non-inverting, LOW = OFF; 12V_PWR supply.
Marcin Kielesiński PROFI-SOFT
Sheet: /
File: ECU.kicad_sch
Title: Fiesta VP37 ECU
Size: User
Date: 2026-11-09
KiCad E.D.A. 9.0.9
Rev: v3.0-PCB1
Id: 1/1